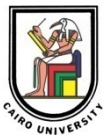
Cairo University
Faculty of science
Mathematics Department
120 minutes.



Final Exam (Summer 2015)
Artificial Intelligence
Comp 408

Answer the following questions. The total credit is 60. **The Exam is in <u>six</u> Pages**

Question (1) (15 points)

A. **(3 points)** How can you define Artificial Intelligence as thinking humanly?

B. **(3 points)** What does it mean for an agent to be omniscient?

C. **(4 points)** Define the PEAS (Performance Measure, Environment, Actuators, Sensors) of an automated taxi driver.

P.T.O.

D. **(6 points)** Explain how does the Genetic Algorithm work? What are the factors that affect its performance?

Question (2) (15 points)

A. **(3 points)** What is the need for Skolem function in First Order Predicate Calculus?

(4 points) Consider the problem of solving one 8-Queens. Give a complete problem formulation. (i.e. description of the state space, initial state, goal state(s), successor function)

B. (**8 points**) Consider the search space below, where *S* is the start node and *G1*, *G2,* and *G3* satisfy the goal test. Arcs are labeled with the cost of traversing them and the estimated cost to a goal is reported inside nodes.

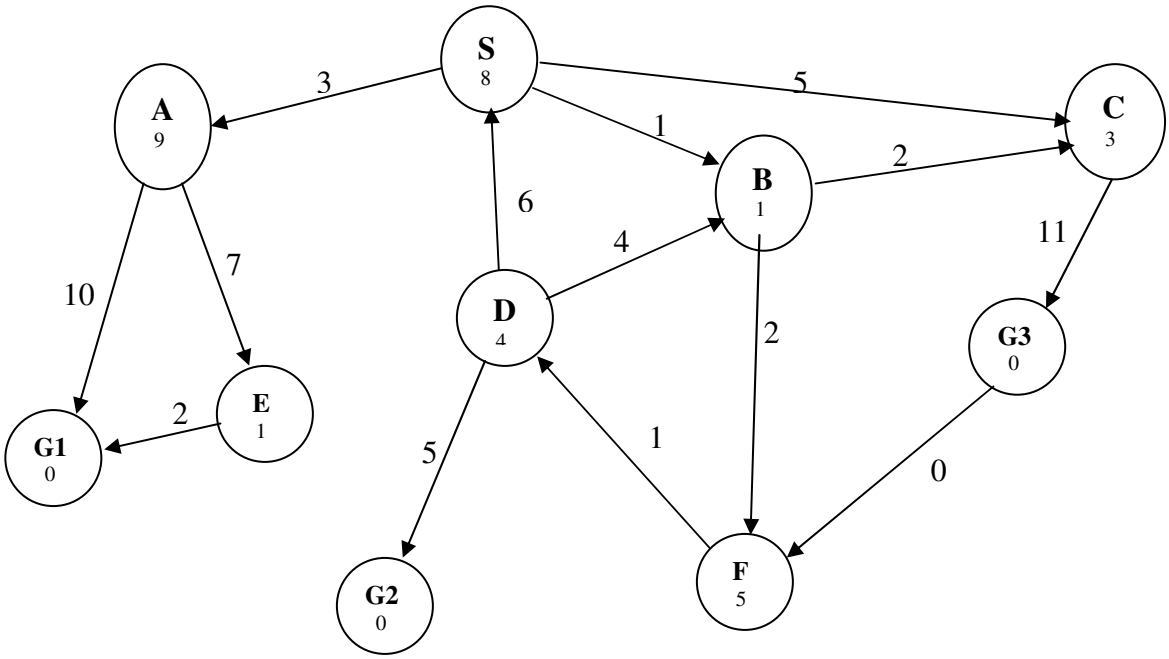
For each of the following search strategies, indicate which goal state is reached (if any) and list, *in order*, and indicate all the states *popped off of the OPEN list*. When all else are equal, nodes should be removed from OPEN in alphabetical order.

i. Breadth First

ii. Depth First

iii. Hill Climbing

iv. A*



Question (3) (15 points)

A. **(6 points)** Just write True/False:

i. (……..) $\exists \mathbf{x}\ \forall y$ equal(x,y) is unsatisfiable

ii. (……..) Depth-first search always expands at least as many nodes as Depth limnited search.

iii. (……..) $(A \leftrightarrow B) \wedge (\neg A \vee B)$ is satisfiable

iv. (……..) Forward Chaining with the Generalized Modus Ponens is complete in First order predicate Calculus.

v. (……..)Prolog works with backward chaining.

vi. (……..)A Unifier is a substitution for variables in atomic sentences that makes them identical

P.T.O.

B. **(10 points)** Consider the sentence "the man walked slowly to the supermarket" and consider the following lexicon and grammars:

Noun –> man | supermarket
V –> walked
Adv –> slowly
Prep –> to
Det –> the

A

S –> NP VP
NP –> Noun
NP –> Article Noun
VP –> VP PP
VP –> VP Adv Adv
VP –> Verb
PP –> Prep NP

B

S –> NP VP
NP –> Noun
NP –> Article NP
VP –> Verb Vmod
Vmod –> Adv Vmod
Vmod –> Adv
Adv –> PP
PP –> Prep NP

C

S –> NP VP
NP –> Noun
NP –> Article NP
VP –> Verb Adv
Adv –> Adv Adv
Adv –> PP
PP –> Prep NP

i. Which of the three grammars, combined with the lexicon, generate the given sentence.
ii. Show the corresponding parse tree.
iii. Write down one sentence of non-English generated by one of the above grammars.
(Explain how it is generated by the grammar)

Question (4) (15 points)

A. **(2 points)** Which of the following pairs can be unified and which can not. For those which can be unified, find a suitable substitution; For those which can not , explain why they cannot be unified.

(Pair a)

(i) likes(homer, dinner(X)) $\wedge$ (today(A) $\rightarrow$ dinner(X)).

(ii) likes(homer, dinner(cooked_by(Y))) $\wedge$ (today(Z) $\rightarrow$ dinner(cooked_by(marge))).

(Pair b)

(i) p(X) $\vee$ (q(f(X)) $\wedge$ r(Y)).

(ii) p(X) $\vee$ (q(X) $\wedge$ r(s)).

P.T.O.

B. **(4 points)** What are the levels of analysis of the natural language.

C. **(2 points)** Which of the following pairs can be unified and which can not. For those which can be unified, find a suitable substitution; For those which can not , explain why they cannot be unified.

(Pair a)

(iii) likes(homer, dinner(X)) $\wedge$ (today(A) $\rightarrow$ dinner(X)).

(iv) likes(homer, dinner(cooked_by(Y))) $\wedge$ (today(Z) $\rightarrow$ dinner(cooked_by(marge))).

(Pair b)

(i) p(X) $\vee$ (q(f(X)) $\wedge$ r(Y)).

(ii) p(X) $\vee$ (q(X) $\wedge$ r(s)).

D. **(4 points)** Translate the following sentences into first-order logic, using a function to represent mother:

i. All dogs are mammals

ii. Fido is a dog

iii. Fido's mother is a mammal

P.T.O.

iv. All mammals have a mother who is a mammal

E. **(3 point)** Convert the following sentence into Conjunction Normal Form
$\forall x([\forall y\ Animal(y) \Rightarrow Loves(x,y)] \Rightarrow [\exists y\ Loves(y,x)])$

F. **(3 point)** Consider the following Knowledge Base
~P(w) v Q(w)
~Q(y) v S(y)
R(z) v S(z)
P(x) v R(x)

Build the refutation resolution tree to prove S(A)

Best Wishes
Areeg Abdalla